



Ethical Hacking Lab with Ettercap

By Albert Zhang

Purpose:

This lab demonstrates how common layer 2 attacks are done utilizing Ettercap and how to prevent set attacks. Performing the attacks yourself is worthwhile because it provides hands-on experience that lets someone thoroughly understand how the attack works. Doing the attack lets someone understand how vulnerabilities are discovered and how an attacker might think.

Background:

In this lab 2 layer 2 attacks are shown, the first one being MAC flooding:

Switches use something called a Mac table that stores Mac addresses of devices and switch ports and switch ports where those devices are connected. Switches use MAC addresses to send data to the correct device instead of broadcasting it to the whole network. A MAC flood attack is when a target's MAC address table (also called a CAM table) is flooded like the name implies, by a huge amount of Ethernet frames. When the MAC table of the switch is filled by the frames generated by the attack (the MAC table of the switch used in this lab holds around 12000 frames), the legitimate MAC addresses could go missing or overwritten, the switch also cannot learn new MAC addresses, these events occurring means the switch is now unable to find destination for existing frames and new frames, making the switch default to flooding all ports expect the incoming one with the frame. When the attacked switch with a full MAC table floods the entire network with the frame it receives the attacker that is connected to network on any given network interface can now receive the frame, originally meant to be sent to a specific.

The Second Attack used in this lab is ARP poisoning:

ARP stands for Address Resolution Protocol, ARP is used when a specific MAC address is needed, so an ARP request is broadcasted to the entire network, saying in simple terms "who has the address of xxx.xxx.x.x?" a device with the address will respond with its MAC address to provide its location. The ARP system is well designed but at the same time it is exploitable, an attacker could "poison" the ARP by sending a malicious packet telling devices that a certain IP's MAC address is the actually the attacker's MAC address, leading to the attackers receiving frames intended for another an IP at a different location, but the malicious packet convinces devices that the attacker's MAC is the correct one. Attackers receiving unintended traffic could lead to usernames and passwords being stolen, files being transferred and more.

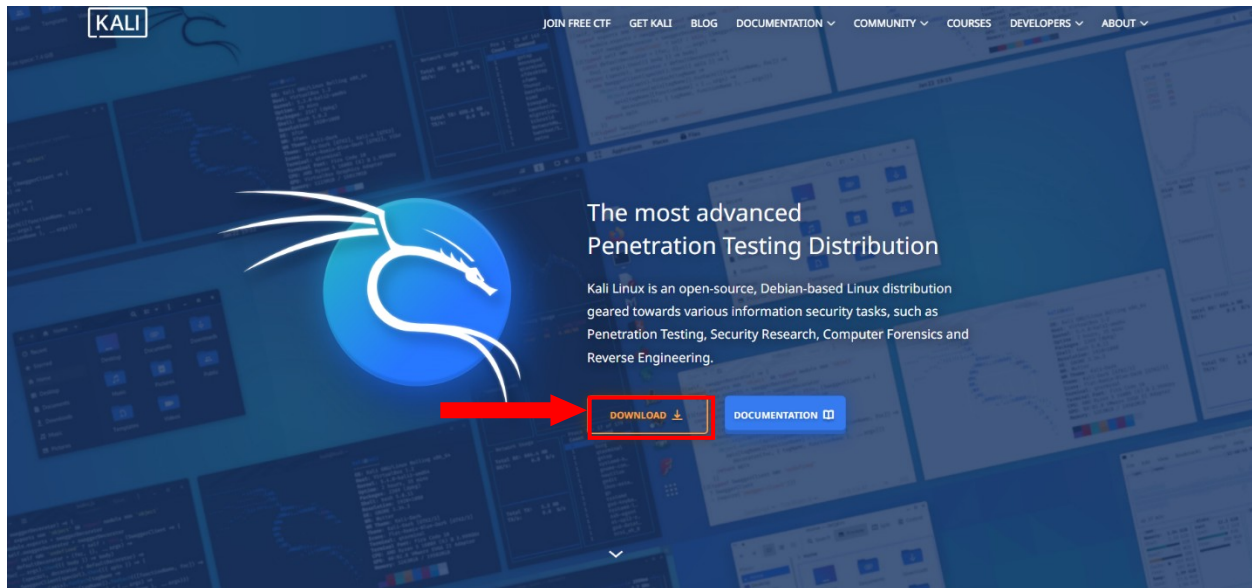
Ettercap: The tools used to perform these attacks in this lab is called Ettercap, Ettercap was developed by Alberto Ornaghi (AloR) and Marco Valleri (NaGaY), two Italian developers. Originally, they were high profile hackers and when authorities found them, and tools they were using, like Ettercap, they weren't arrested instead the authorities wanted to hire them because of their skills in hacking.

Later Ettercap was released to the public, becoming an open-source network security tool that can perform various attacks easily, used to test network security. Ettercap requires Kali Linux, which is very popular in cybersecurity, with many pre-installed networking, though using these tools and Ettercap is only legal in an ethical testing environment, for educational purposes.

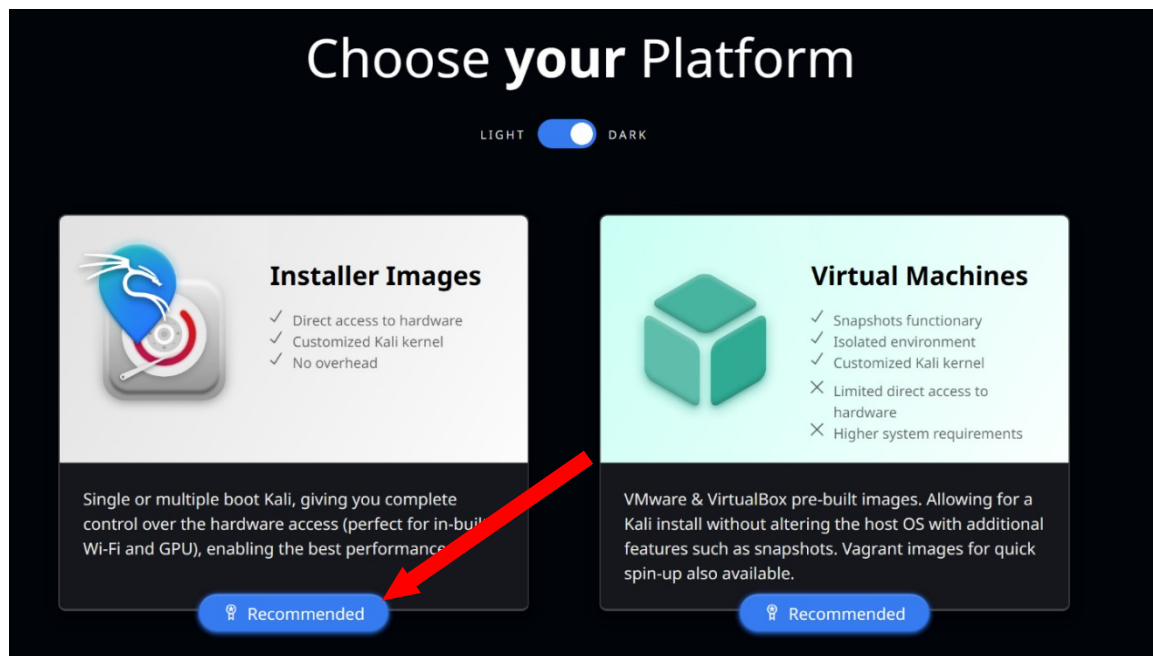
Configurations:

Installing Kali Linux:

Go to <https://www.kali.org/> and click on download.

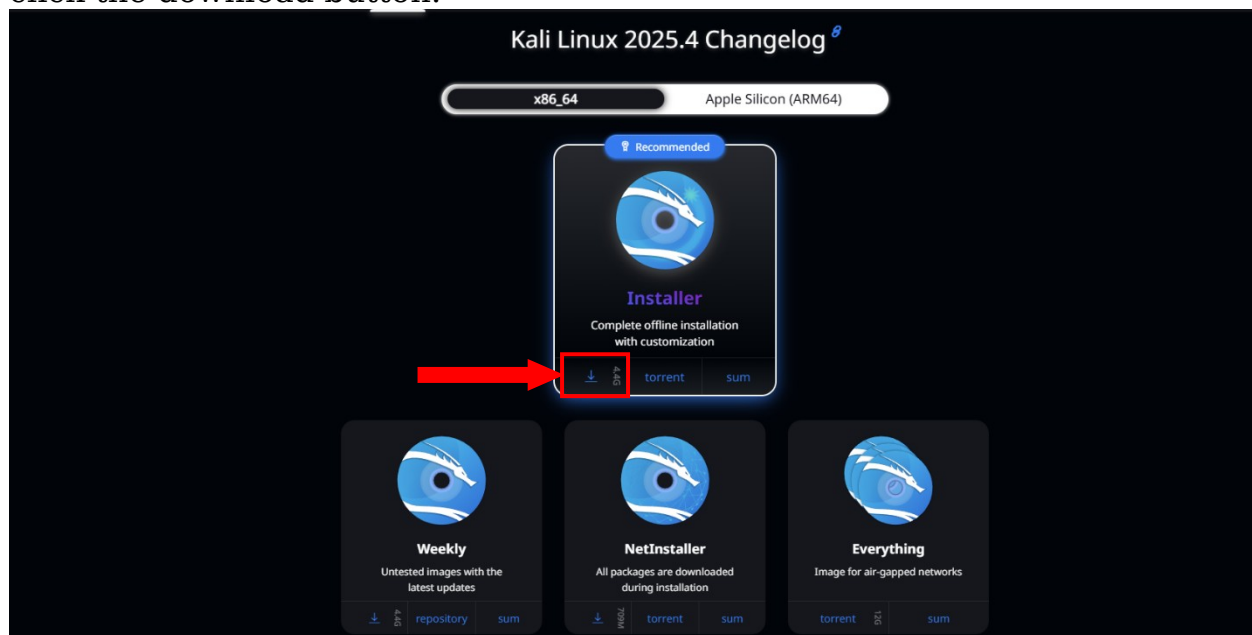


Select either option to choose between installing an ISO image, or a virtual



machine.

In this lab an ISO installation is used, select the recommended installer and click the download button.



Next, go to <https://rufus.ie/> and download to rufus dependency for the installation of the ISO image, select the portable type (second option below).



Download

Latest releases:

Link	Type	Platform	Size	Date
rufus-4.13.exe	Standard	Windows x64	1.9 MB	2026.02.17
rufus-4.13p.exe	Portable	Windows x64	1.9 MB	2026.02.17
rufus-4.13_x86.exe	Standard	Windows x86	1.8 MB	2026.02.17
rufus-4.13_arm64.exe	Standard	Windows ARM64	4.9 MB	2026.02.17

[Other versions \(GitHub\)](#)

[Other versions \(dAppCDN\)](#)

Warning: Copy the settings below onto your drive and run the rufus-4.13p.exe, this **WILL WIPE** your drive, so backup or transfer wanted files

Drive Properties

Device
NO_LABEL (I:) [256 GB]

Boot selection
kali-linux-2024.3-installer-amd64.iso ☒ SELECT

Persistent partition size
 0 (No persistence)

Partition scheme
MBR

Target system
BIOS or UEFI

▼ Show advanced drive properties

Format Options

Volume label
Kali Linux amd64 1

File system
Large FAT32 (Default)

Cluster size
32 kilobytes (Default)

▼ Show advanced format options

Status

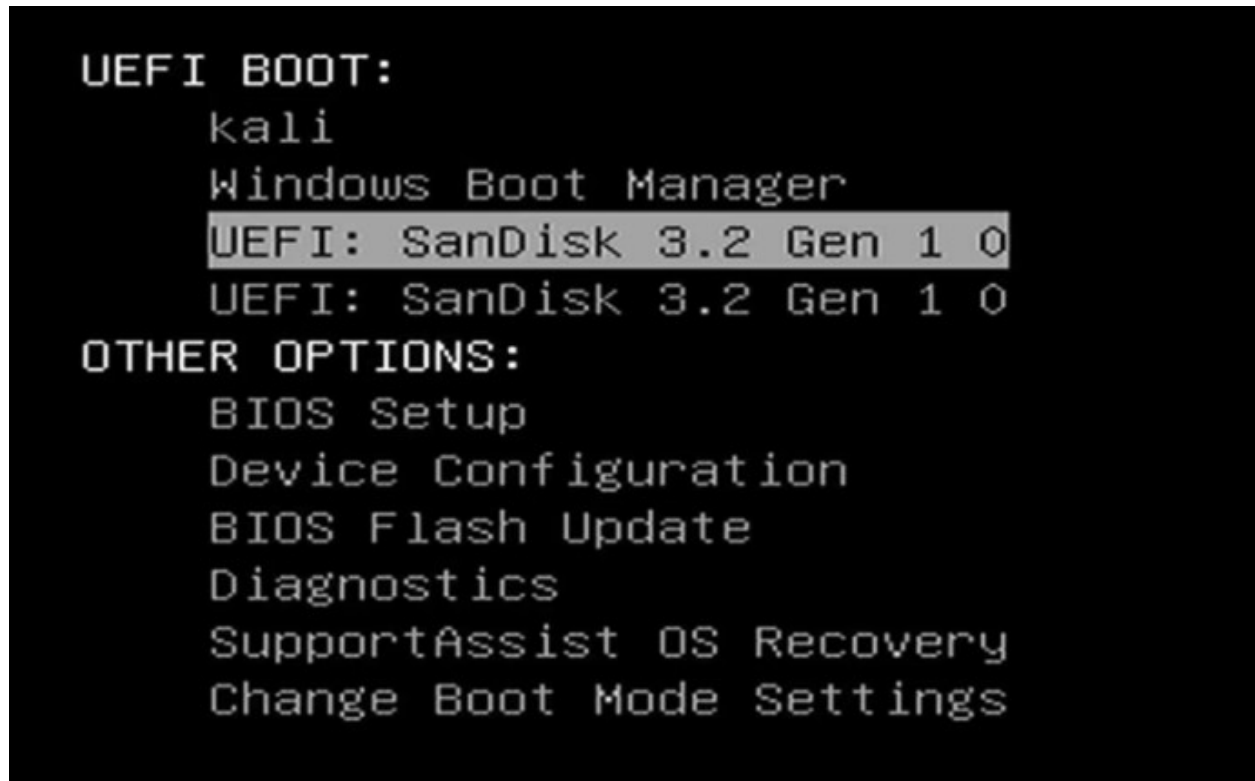
Deleting partitions (this may take a while)...

Checking for conflicting processes... 00:00:00

After, reboot your PC without touching the USB drive, enter boot mode by press tab, ESC and F1-F12, or enter boot mode during bootup.

Select the USB drive used with arrow keys and then press enter.



After Clicking enter, your screen will look like this:



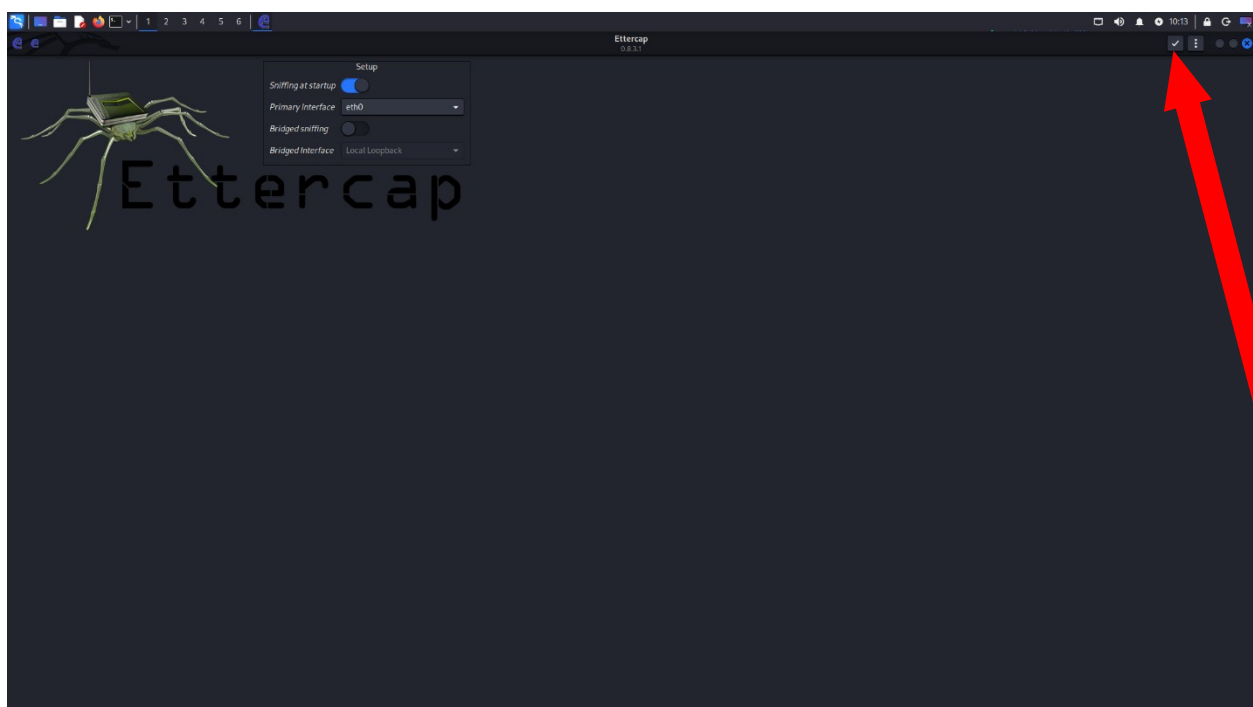
Next you will be prompted to setup Kali Linux, enter your login and click "Log In", after that you will have successfully finished installing Kali Linux

Using Ettercap:

Open Ettercap with console with the command “sudo Ettercap -G”

```
Session Actions Edit View Help
(antigua@AntiguaHost)-[~] youtube.com/watch?v=M8C...UE88
$
(antigua@AntiguaHost)-[~]
$
(antigua@AntiguaHost)-[~]
$
(antigua@AntiguaHost)-[~]
$ sudo etter
[sudo] password for antigua:
sudo: etter: command not found
(antigua@AntiguaHost)-[~]
$ sudo ettercap -G
Ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team
```

Once Ettercap is open click the checkmark on the upper right corner of the menu to start unified sniffing

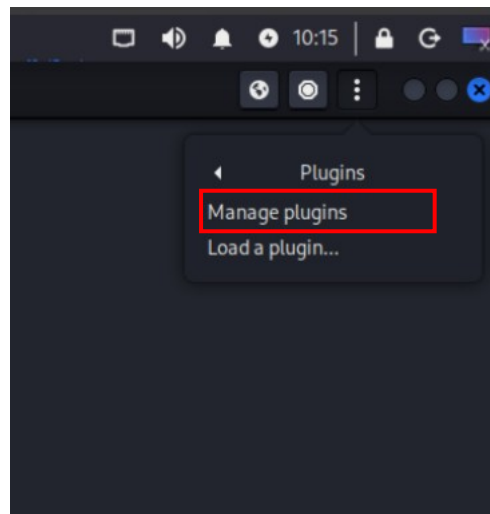
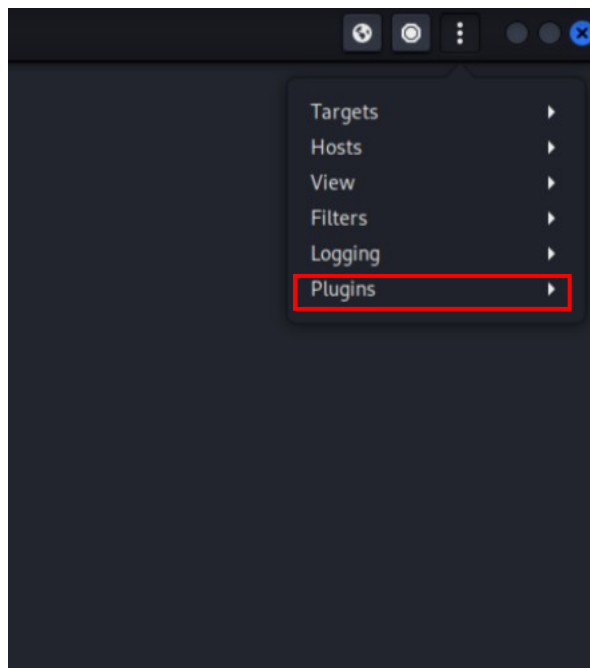


This message will appear in the console at the bottom of the menu once the sniffing starts working.

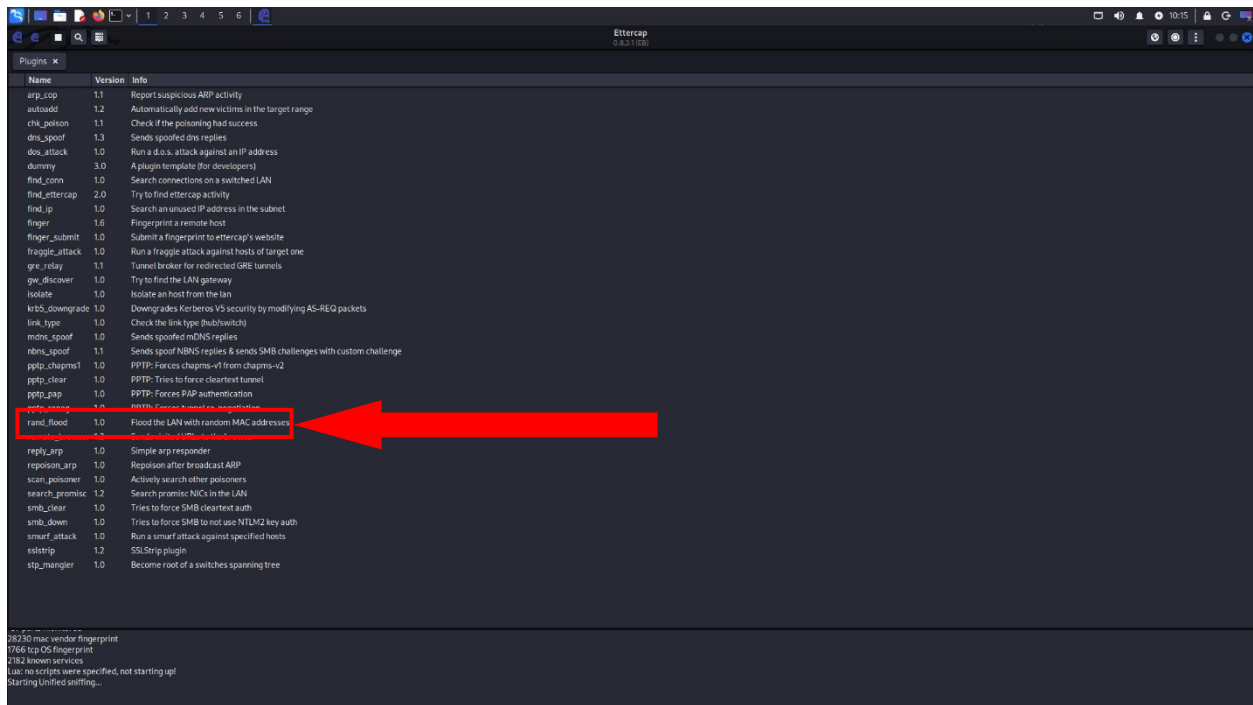
```
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!
Starting Unified sniffing...
```

MAC Flooding:

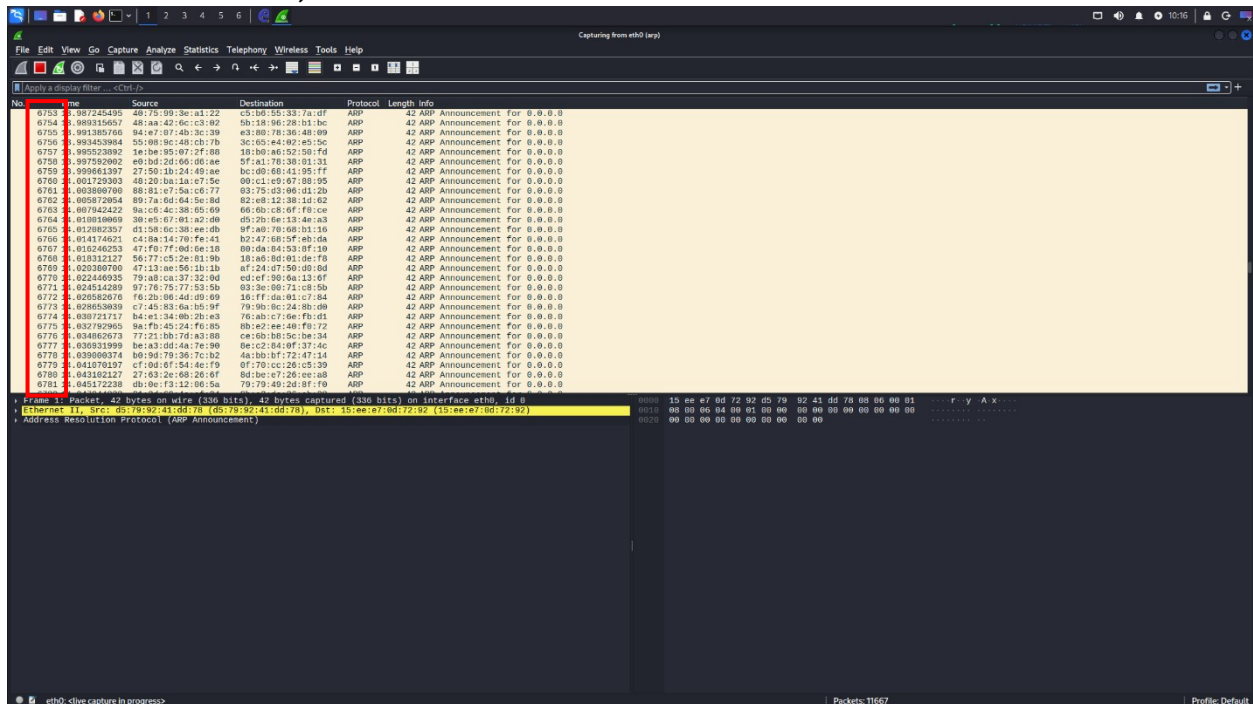
Click on the 3 dots, and then “Plugins”, after that select “Manage plugins



Now you will have a list of all the available attacks that Ettercap offers, in this case run “rand_flood” to start a MAC flooding attack.

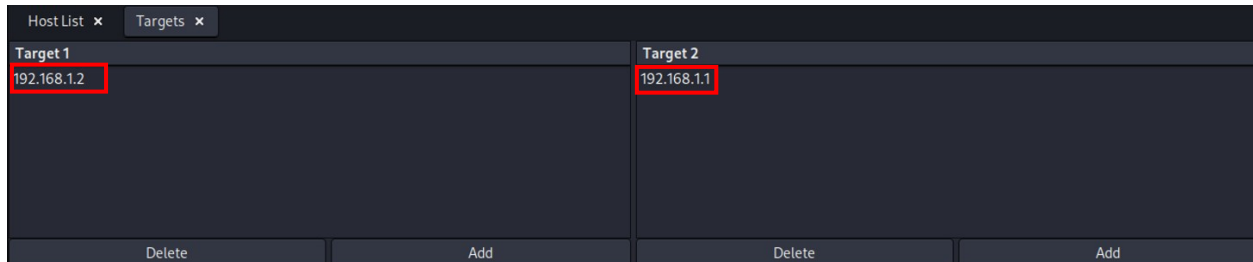


After stating rand_flood, thousands of MAC addresses will be forwarded in a matter of seconds, shown in Wireshark

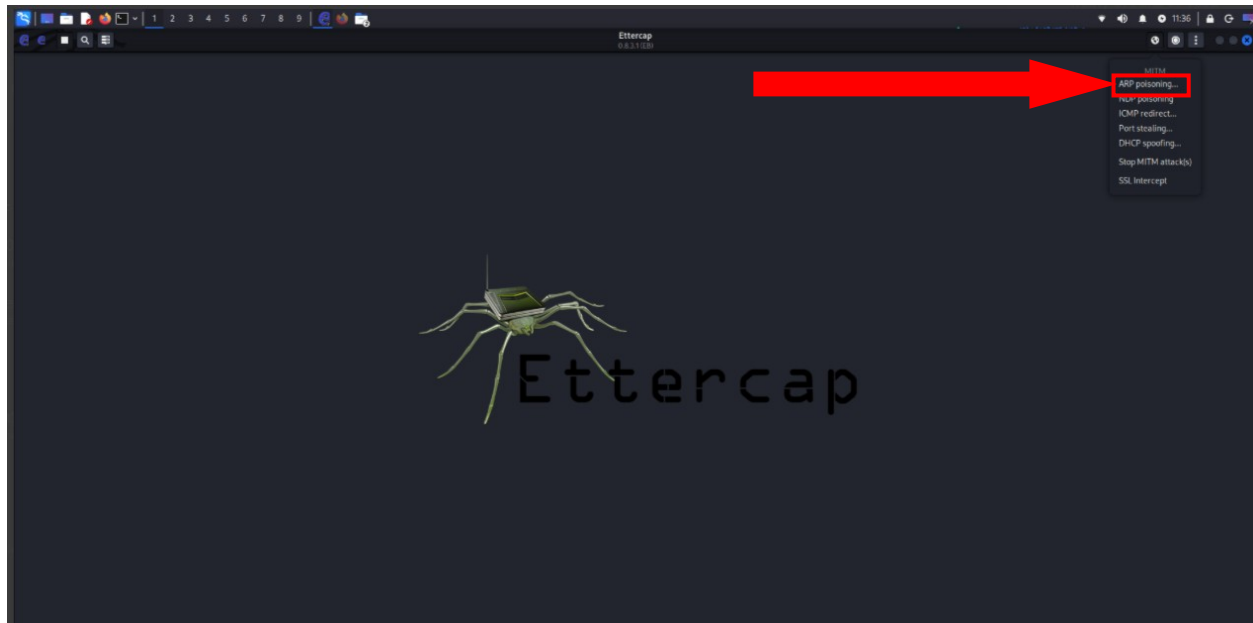


ARP poisoning:

First a network is set up, where the addresses 192.168.1.1 and 192.168.1.2 is used as the router and another PC in the network.

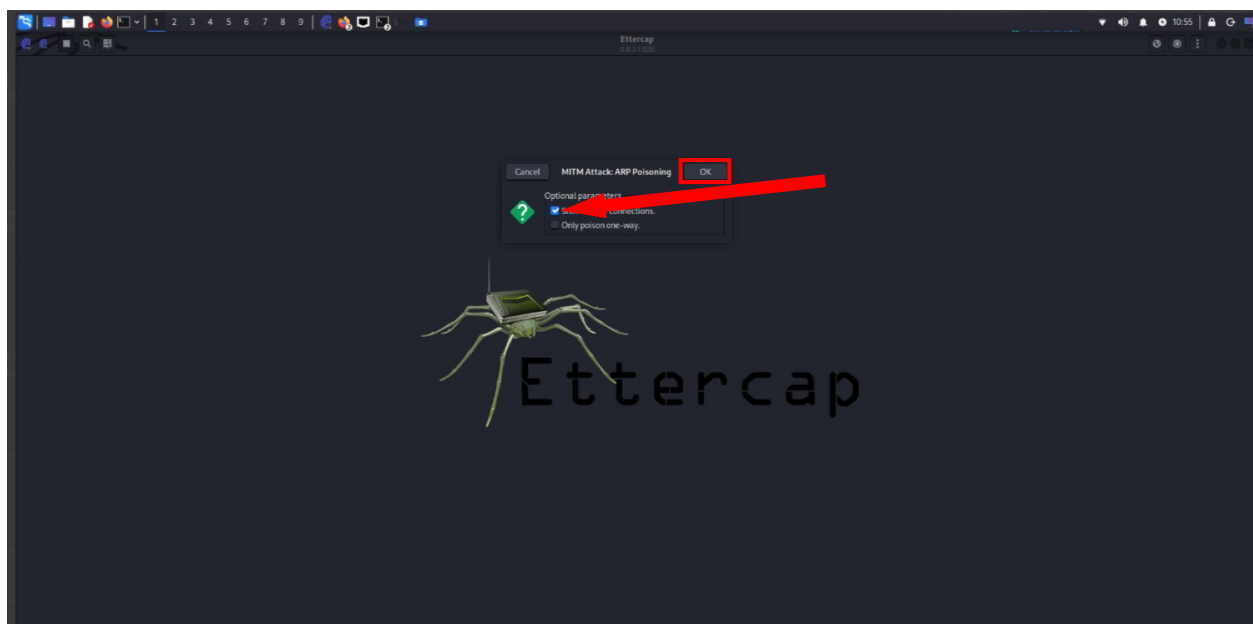


Ettercap is going to locate these 2 hosts, after click on the 3 dots again, and



select "ARP poisoning".

Enable sniff remote connections and then click "OK".



After clicking “OK” in the terminal bellow, you will see the poisoning start working:

```
o1 DHCP: [CC:98:91:B0:AE:60] DISCOVER
Randomizing 255 hosts for scanning...
y1 Scanning the whole netmask for 255 hosts...
DHCP: [CC:98:91:B0:AE:60] DISCOVER
e 3 hosts added to the hosts list...
y DHCP: [CC:98:91:B0:AE:60] DISCOVER
e DHCP: [CC:98:91:B0:AE:60] DISCOVER
e DHCP: [CC:98:91:B0:AE:60] DISCOVER
F DHCP: [CC:98:91:B0:AE:60] DISCOVER
T DHCP: [CC:98:91:B0:AE:60] DISCOVER
d DHCP: [CC:98:91:B0:AE:60] DISCOVER
T DHCP: [CC:98:91:B0:AE:60] DISCOVER
T DHCP: [CC:98:91:B0:AE:60] DISCOVER
a1 DHCP: [CC:98:91:B0:AE:60] DISCOVER
e DHCP: [CC:98:91:B0:AE:60] DISCOVER
DHCP: [CC:98:91:B0:AE:60] DISCOVER
DHCP: [CC:98:91:B0:AE:60] DISCOVER
Host 192.168.1.2 added to TARGET1
Host 192.168.1.1 added to TARGET2

ARP poisoning victims:
n GROUP 1 : 192.168.1.2 04:D9:C8:BA:26:46
GROUP 2 : 192.168.1.1 DC:F7:19:78:48:90
```

Back in the “Manage Plugins” menu there is a `chk_posion` tool that you can run, and it will report if the poisoning of the ARP was successful.

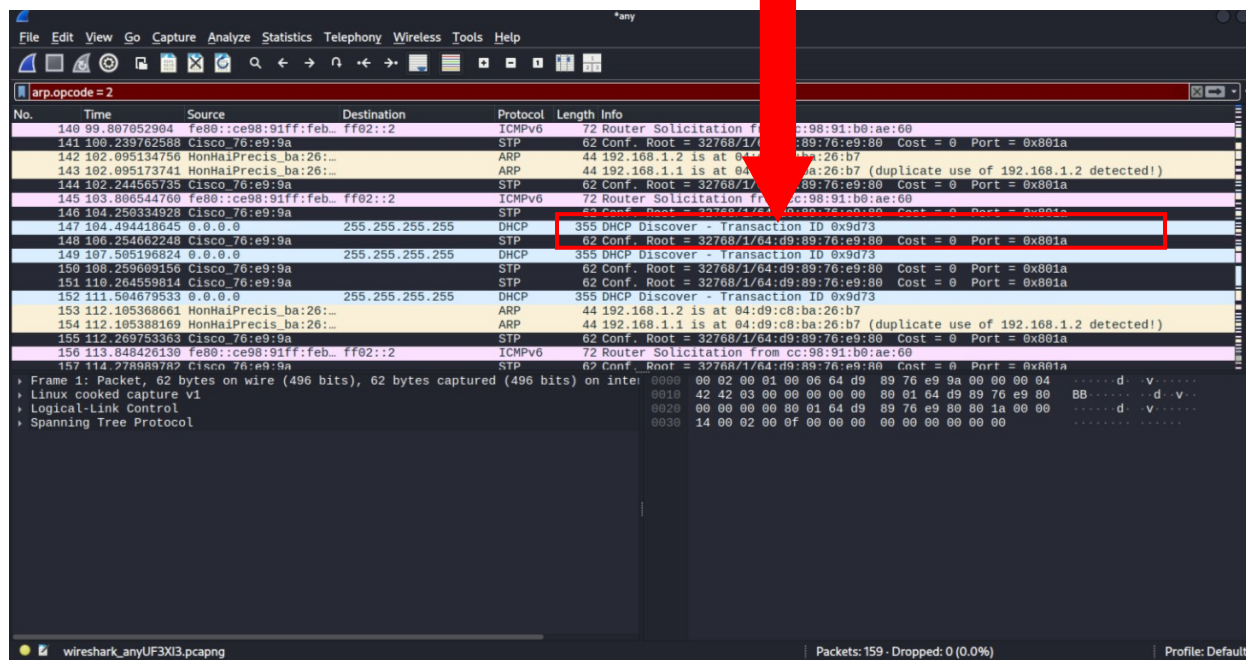
```
HCP: [CC:98:91:B0:AE:60] DISCOVER
andomizing 255 hosts for scanning...
anning the whole netmask for 255 hosts...
HCP: [CC:98:91:B0:AE:60] DISCOVER
hosts added to the hosts list...
HCP: [CC:98:91:B0:AE:60] DISCOVER
HCP: [CC:98:91:B0:AE:60] DISCOVER
ost 192.168.1.1 added to TARGET1
ost 192.168.1.2 added to TARGET2

RP poisoning victims:

ROUP 1 : 192.168.1.1 DC:F7:19:78:48:90

ROUP 2 : 192.168.1.2 04:D9:C8:BA:26:46
ctivating chk_poison plugin...
k_poison: Checking poisoning status...
k_poison: Poisoning process successful!
```

You can also use Wireshark and see that duplicate address are being detected, meaning the attacker is trying to impersonate other addresses on the network to receive their traffic.



Port Security:

Without port security the MAC address of a router could look like:

```
Switch#show mac address-table
Mac Address Table
-----
```

Vlan	Mac Address	Type	Ports
All	0100.0ccc.cccc	STATIC	CPU
All	0100.0ccc.cccd	STATIC	CPU
All	0180.c200.0000	STATIC	CPU
All	0180.c200.0001	STATIC	CPU
All	0180.c200.0002	STATIC	CPU
All	0180.c200.0003	STATIC	CPU
All	0180.c200.0004	STATIC	CPU
All	0180.c200.0005	STATIC	CPU
All	0180.c200.0006	STATIC	CPU
All	0180.c200.0007	STATIC	CPU
All	0180.c200.0008	STATIC	CPU
All	0180.c200.0009	STATIC	CPU
All	0180.c200.000a	STATIC	CPU
All	0180.c200.000b	STATIC	CPU
All	0180.c200.000c	STATIC	CPU
All	0180.c200.000d	STATIC	CPU
All	0180.c200.000e	STATIC	CPU
All	0180.c200.000f	STATIC	CPU
All	0180.c200.0010	STATIC	CPU
All	ffff.ffff.ffff	STATIC	CPU
1	0010.a36e.e760	DYNAMIC	Fal/0/24
1	0012.a34f.7ce2	DYNAMIC	Fal/0/24
1	001b.ee63.9b05	DYNAMIC	Fal/0/24
1	0029.a507.33f7	DYNAMIC	Fal/0/24
1	003a.dc60.0c4b	DYNAMIC	Fal/0/24
1	0040.4f57.c39e	DYNAMIC	Fal/0/24
1	0045.a94f.d893	DYNAMIC	Fal/0/24
1	0054.3f23.2ed7	DYNAMIC	Fal/0/24
1	0055.e702.f1d0	DYNAMIC	Fal/0/24
1	0059.e06c.c0bb	DYNAMIC	Fal/0/24
1	006c.8612.f9de	DYNAMIC	Fal/0/24
1	006f.ef6a.16a6	DYNAMIC	Fal/0/24
1	0078.d838.edd8	DYNAMIC	Fal/0/24
1	0081.0846.aad9	DYNAMIC	Fal/0/24
1	0081.9315.ca1c	DYNAMIC	Fal/0/24
1	008d.633e.345f	DYNAMIC	Fal/0/24
1	0092.7217.d184	DYNAMIC	Fal/0/24
1	0098.d15d.a197	DYNAMIC	Fal/0/24
1	009f.3e2a.bc97	DYNAMIC	Fal/0/24
1	00a5.9e06.9115	DYNAMIC	Fal/0/24
1	00ab.3d4b.a358	DYNAMIC	Fal/0/24
1	00af.641a.018b	DYNAMIC	Fal/0/24
1	00ba.5955.155c	DYNAMIC	Fal/0/24
1	00ba.eb3e.6a1c	DYNAMIC	Fal/0/24
1	00bb.cf58.348a	DYNAMIC	Fal/0/24
1	00bf.be75.3b81	DYNAMIC	Fal/0/24
1	00c4.6b4f.fac6	DYNAMIC	Fal/0/24
1	00c4.985e.f150	DYNAMIC	Fal/0/24
1	00d2.7d61.924d	DYNAMIC	Fal/0/24
1	00d4.1d5c.7476	DYNAMIC	Fal/0/24
1	00ee.7a55.6662	DYNAMIC	Fal/0/24
1	0203.5263.b8fc	DYNAMIC	Fal/0/24
1	0203.952b.327e	DYNAMIC	Fal/0/24
1	0206.c064.76d0	DYNAMIC	Fal/0/24
1	0210.4919.95fe	DYNAMIC	Fal/0/24
1	0219.a047.6653	DYNAMIC	Fal/0/24
1	0229.f444.51c6	DYNAMIC	Fal/0/24
1	022a.2e75.58b6	DYNAMIC	Fal/0/24
1	023c.8e75.b773	DYNAMIC	Fal/0/24
1	023d.7535.a5df	DYNAMIC	Fal/0/24
1	024d.034f.3872	DYNAMIC	Fal/0/24
1	0251.0567.ce35	DYNAMIC	Fal/0/24
1	0254.266f.9bee	DYNAMIC	Fal/0/24
1	0254.5244.f576	DYNAMIC	Fal/0/24
1	0254.7b2c.cce8	DYNAMIC	Fal/0/24
1	0264.ac09.2396	DYNAMIC	Fal/0/24
1	0265.f54e.e409	DYNAMIC	Fal/0/24
1	027b.cf20.3e6f	DYNAMIC	Fal/0/24
1	027d.5d32.5140	DYNAMIC	Fal/0/24
1	0295.621b.8e79	DYNAMIC	Fal/0/24
1	0296.604e.2a91	DYNAMIC	Fal/0/24

This screenshot shows how the entire MAC table is full, making the switch default back to flooding all ports, port security prevents this

Configuration of Port Security:

```
Switch#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#int range f1/0/##
```

```
Switch(config-if-range)#switchport mode access
```

```
Switch(config-if-range)#switchport port-security maximum 2
```

```
Switch(config-if-range)#exit
```

```
Switch(config)#int range f1/0/##
```

```
Switch(config-if-range)#switchport mode access
```

```
Switch(config-if-range)#switchport port-security
Switch(config-if-range)#switchport port-security maximum 2
Switch(config-if-range)#switchport port-security mac-address
sticky
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#exit
Switch#
```

After issuing these commands on the switch, only 2 mac addresses will be present at a time and the switch will also dynamically learn the addresses, if you have configured port security correctly, you should verify by doing show run, with your interfaces looking like this:

```
interface FastEthernet1/0/1
switchport mode access
switchport port-security maximum 2
switchport port-security
switchport port-security mac-address sticky
!
interface FastEthernet1/0/2
switchport mode access
switchport port-security maximum 2
switchport port-security
switchport port-security mac-address sticky
```

Problems:

First problem we faced was running Ettercap, where Kali Linux had to be used, so we figure out the process of installing Kali Linux, then learning how the operating system worked, such as simple things like taking a screenshot and we also tried to install many tools such as Ettercap, and Wireshark before finding out they came preinstalled with Kali Linux.

Another problem we faced while Kali Linux was the constant crashing, even though Kali Linux came preinstalled with many networking tools, most performance dependencies and drivers weren't, which we had to install.

Conclusion:

Ethical hacking is valuable and important in a testing environment because it provides someone trying to make a network secure an attacker's perspective of how the attacks work, and gives insight on how an attacker might think, letting the network engineer mitigate these attacks efficiently by understanding how vulnerabilities exist, though tools used to attack or eavesdrop a network are only legal to use in a testing environment setup only for educational purposes.